

INDU 211 · Rapid Review Sheet (1-Page Cheatsheet)

Part 2: Chapter 3 — Manufacturing & Process Engineering

1. THE DESIGN-MANUFACTURING INTERFACE

- Core Friction:** Product Designer wants *tight tolerances* (high cost) $\longleftrightarrow$ Manufacturing Engineer wants *loose tolerances* (low cost, easy tooling).
- Concurrent Engineering:** Simultaneous integration of design, manufacturing, testing, maintenance, and recycling from Day 1. Eliminates "throwing design over the wall"; pioneered in aerospace.
- Bill of Materials (BOM):** Inverted hierarchical tree: Product (L0) $\rightarrow$ Sub-assemblies (L1) $\rightarrow$ Components $\rightarrow$ Raw Materials.

2. QUANTITATIVE ECONOMICS & BREAK-EVEN FORMULAS

- Linear Cost Model:** $TC(X) = a \cdot X + b$ where b = Fixed Cost (FC), a = Variable Cost/unit (VC), X = volume.
- Single-Process Break-Even:** $X_{BEP} = \frac{FC}{P-a}$ where P = Selling Price/unit, $(P - a)$ = Contribution Margin.
- Multi-Process Crossover Volume:** Point where two processes cost the same:

$$FC_1 + a_1X = FC_2 + a_2X \implies X_{\text{crossover}} = \frac{FC_1 - FC_2}{a_2 - a_1}$$

- Decision Golden Rule:**
 - Low Volume:** Select process with **lowest Fixed Cost** (e.g., manual machining or 3D printing).
 - High Volume:** Select process with **lowest Variable Cost** (e.g., automated die stamping / casting).

3. OPERATION SEQUENCING RULES & STEEL SHAFT FLOW

- 3 Rules:** 1. Shortest path (no backtracking) | 2. Non-interference (subsequent ops don't ruin finishes/burrs) | 3. Datum preservation (group cuts per machine setup).
- Steel Shaft Sequence:**

Cut Bar Stock (Bandsaw) $\rightarrow$ Face Ends (Lathe) $\rightarrow$ Turn Diameters (Lathe) $\rightarrow$ Drill Holes (Lathe/Press)

$\rightarrow$ Groove Undercuts (Lathe) $\rightarrow$ Heat Treatment (Furnace) $\rightarrow$ Cylindrical Grinding (Precision) $\rightarrow$ Protective Coating

4. MASTER MANUFACTURING PROCESSES REFERENCE TABLE

Process Family	Method	Core Characteristics	Industrial Applications
Casting	Molten liquid poured into mold cavity.	Sand: Low tooling cost, rough surface, expendable mold. Permanent / Die: High tooling cost, smooth, high volume.	Engine blocks, pump housings, brackets, large machinery frames.
Hot Working	Formed ABOVE recrystallization temp ($T > 0.5T_m$).	Low forces required; large deformations; rough scale surface.	Hot rolling, initial ingot breakdown, heavy railway rails, I-beams.
Cold Working	Formed BELOW recrystallization temp (room temp).	Strain hardening (stronger/harder); high forces; close tolerances.	Cold rolling, cold wire drawing, automotive body panels.
Forging	Intermittent impact hammering or continuous press.	Aligns internal grain flow with contours; high toughness.	Engine crankshafts, connecting rods, aircraft landing gear.
Extrusion	Compressed & forced through a shaped die orifice.	Constant cross-section; "toothpaste tube" mechanics.	Aluminum window frames, structural tubes, architectural trim.
Wire Drawing	Rod pulled in tension through conical die.	Reduces cross-sectional diameter; increases tensile strength.	Electrical copper wire, structural cable, steel wire ropes.
Sheet Metal	Shearing, bending, or deep drawing sheet.	Punching: cut hole is scrap. Blanking: cut piece is part. Deep Drawing: seamless hollow vessels (cans, sinks).	Beverage cans, kitchen sinks, automotive hoods and fenders.
Turning	Lathe; workpiece rotates , single-point tool feeds.	Cylindrical profiles, tapers, external threads, shoulders.	Shafts, pins, axles, threaded studs, precision bushings.
Milling	Mill; cutter rotates , multi-tooth tool, part feeds.	High versatility: flat faces, keyway slots, pockets, 3D contours.	Engine heads, molds, brackets, transmission casings.
Shaping vs Planing	Reciprocating single-point flat cutting.	Shaping: Workpiece stationary; tool moves (smaller parts). Planing: Tool stationary; heavy workpiece moves (huge beds).	Machine tool guide beds, railway track switches, flat slides.
Broaching	Multi-tooth straight bar pushed/pulled in 1 pass .	Extremely fast; tight tolerances; expensive custom tool.	Internal splines, keyway slots, square/hexagonal holes.
Grinding	Rotating bonded abrasive wheel; microscopic chips.	Machining post-heat-treat hardened metals; mirror finish.	Bearing races, ground shafts, cutting tool sharpening.